

ABSTRACT

This project research was designed and implemented to address the growing challenge of email spam detection using a hybrid feature-selection technique integrated with a Deep Neural Network (DNN) classifier. Email remains a primary communication channel, but the growing volume of spam poses severe security and productivity risks. Traditional models often struggle with high-dimensional features generated during text vectorisation, leading to computational inefficiency and overfitting. Using a balanced dataset of 5,000 labelled Enron emails sourced from Kaggle (2,500 spam, 2,500 ham), the raw text was preprocessed and converted into a 25,000-dimensional TF-IDF feature matrix. A two-stage hybrid pipeline combining a Chi-Square statistical filter with an L1-regularised embedded selector reduced this space by approximately 99 percent, isolating the 251 most discriminative features.

The selected features were used to train a DNN classifier comprising three fully connected hidden layers with 256, 128, and 64 neurons respectively, each followed by Batch Normalisation and Dropout to control overfitting, with a sigmoid output neuron for binary classification. Training used the Adam optimiser with early stopping and a learning rate scheduler on an 80/20 stratified split. Evaluated on 1,000 unseen test emails, the model achieved an accuracy of 96.30%, precision of 95.48%, recall of 97.20%, an F1-score of 96.33%, and a ROC-AUC of 99.03%.

The system was deployed as a functional REST API and web application, demonstrating practical viability beyond a theoretical exercise. In summary, the hybrid approach proved effective in reducing dimensionality without sacrificing accuracy. The study recommends evaluating the model on larger, more recent datasets, incorporating periodic retraining to track evolving spam patterns, and exploring alternative hybrid feature-selection combinations with more advanced architectures in future work.